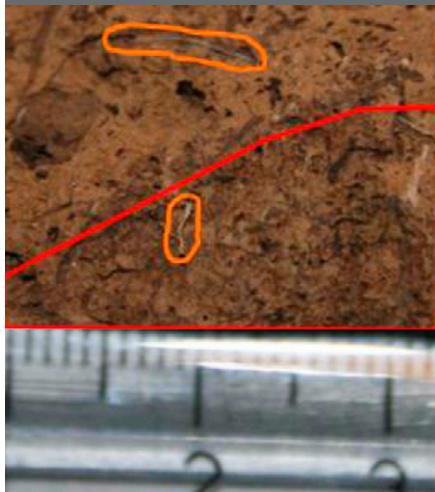


Special Section:
Preferential Flow

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The two-dimensional spatial distribution of soil organic matter composition at intact surfaces of preferential flow paths was analyzed with Fourier transformed infrared spectroscopy in diffuse reflectance mode. Resulting millimeter-scale maps of the relation between CH and CO functional groups allowed distinguishing earthworm burrows, root channels, and aggregate coatings.

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DRIFT Mapping of Organic Matter Composition at Intact Soil Aggregate Surfaces

In structured soils, interaction of percolating water and reactive solutes with the soil matrix is often restricted to the outer surfaces of the preferential flow paths. Such surfaces of soil aggregates and biopores are mostly covered by organic matter (OM) that finally controls wettability, sorption, and transfer properties of the flow pathways. However, the local OM properties along such surfaces are largely unknown to date because the coatings are relatively thin and vulnerable. The objective of this study was to determine and compare the local two-dimensional distribution of soil OM composition at intact aggregate surfaces that serve as preferential flow paths. The Fourier transformed infrared spectroscopy in diffuse reflectance mode (diffuse reflectance infrared Fourier transform, DRIFT) was applied to determine transects and grids of OM functional group data (i.e., CH/CO ratios) on undisturbed and intact surfaces of soil aggregate samples using a DRIFT mapping procedure in 1-mm steps. The aggregate sample surfaces could be distinguished by DRIFT mapping in areas from earthworm burrows, root channels, and aggregate coatings. The water drop infiltration time of these structural surfaces appeared to correspond with CH/CO ratios for uncoated crack surfaces but less so for earthworm burrows. The results show that coatings at preferential flow-path surfaces differed locally in terms of OM composition, distribution, and possibly also in wettability, indicating yet unknown implications for preferential movement of water and reactive solutes.

Abbreviations: A, absorption bands of C–H stretch vibrations; Ar, arable; B, absorption bands of C=O stretch vibrations; CH, alkyl groups; C=O, carboxyl and carbonyl groups; DRIFT, diffuse reflectance infrared Fourier transform; G, grassland; KM, Kubelka–Munk; OH, hydroxyl groups; OM, organic matter; SOM, soil organic matter.

In structured soils, preferential flow may occur along earthworm burrows, root channels, or the interaggregate pore space (e.g., Beven and Germann, 1982; Ghodrati and Jury, 1992; Flury et al., 1994; Nieber, 2000). The process is highly dynamic and mostly temporally restricted to times during and after storm events. When moving through such pathways, the percolating water is bypassing much of the volume of the porous soil matrix (e.g., Jarvis, 2007; Köhne et al., 2009a). Since percolating water includes dissolved chemicals, colloids, and dispersed particles, preferential transport of solutes or cotransport of contaminants may potentially be hazardous for groundwater and surface water. The rapid movement through a small fraction of the soil volume strongly limits the soil's filter and buffer function (e.g., Köhne et al., 2009b). Moreover, any contact and exchange between the preferentially moving water and the surrounding soil matrix is largely restricted to the surface areas of such flow paths. As for mineral particles, the surfaces of porous soil aggregates are mostly covered by OM. Thus, the OM composition at these surfaces may control wettability and sorption and transfer properties of flow pathways.

The soil organic matter (SOM) is composed of a complex mixture of organic components that are different in composition (i.e., the chemical structure) and conformation (i.e., the spatial molecular structure), among other properties (e.g., Hayes and Clapp, 2001; Negre et al., 2002). The OM characteristics affect solubility, wettability, stability, and sorptivity (e.g., Guggenberger et al., 1996; Capriel et al., 1995; Capriel, 1997). The SOM has frequently been fractionated by physical and chemical procedures from mostly disturbed bulk soil samples (Hayes and Clapp, 2001; Kaiser et al., 2002; Ludwig et al., 2003).

Spectroscopic techniques such as transmission FTIR allow analyzing the composition functional groups in bulk soil OM (e.g., MacCarthy and Rice, 1985; Kögel-Knabner, 1997; Celi et al., 1997; Capriel, 1997) as well as of SOM fractions separated or extracted from samples (e.g., Gerzabek et al., 1997; Tatzber et al., 2007; Kaiser et al., 2007). Diffuse reflection infrared Fourier Transform (DRIFT) spectroscopy in the mid infrared region

(2.5–25 μm wavelengths) has also been used to characterize SOM composition (e.g., Haberhauer and Gerzabek, 1999; Reeves et al., 2001) or to determine C contents in soil (e.g. Reeves, 2003; Reeves et al., 2005).

The absorption intensities of bands A_a (asymmetric stretch) and A_s (symmetric stretch) (i.e., at wave numbers of about 2930 and 2855 cm^{-1}) indicate C–H stretch vibrations of alkyl groups and are assumed to reflect the hydrophobic properties of OM (Capriel et al., 1995) since alkyl groups show a hydrophobic behavior (Morrison and Boyd, 1983). These bands are usually superimposed as a shoulder of the broad OH band located at 3750 to 2700 cm^{-1} (Hesse et al., 1984). The absorption bands B (i.e., summarized heights of the absorption bands at wave numbers ranging from 1740 to 1698 cm^{-1} and from 1640 to 1600 cm^{-1}) are caused by stretch vibrations of C=O groups of carboxylic acids and ketones (Van der Marel and Beutelspacher, 1976) or carboxylate anions (Celi et al., 1997) and reflect relatively hydrophilic OM functional groups (Morrison and Boyd, 1983).

The contents of C=O groups in SOM determined by using FTIR spectroscopy are proportional to their cation exchange capacity (Celi et al., 1997; Kaiser et al., 2008), while the ratio between hydrophobic (A) and hydrophilic (B) groups could be used to describe the potential wettability of soils (e.g., Ma'shum and Farmer, 1985; Hayes and Clapp, 2001; Ellerbrock et al., 2005). The properties and OM composition of soil from aggregate coatings was found to be different from that of the interior parts of aggregates (Ellerbrock and Gerke, 2004; Urbanek et al., 2007). When SOM fractions are obtained destructively (i.e., either by chemical fractionation or by peeling off shells from aggregates), their composition may not be identical with the SOM at aggregate surfaces where complexes with soil minerals or polyvalent cations dominate (e.g., Ziehmman, 1980; Sposito, 1989; von Lützow et al., 2007).

Other in situ methods, such as FTIR microscopy (Gregoriou and Rodman, 2002; Aparicio et al., 2002), that use intact samples embedded in poly-resins and thin sections allow a spatially resolved analysis of surfaces (Mertig et al., 2002) and a spatial allocation of functional characteristics of organic components to micromorphological structures (Zieba-Palus, 1997; Aparicio et al., 2002; He et al., 2002). However, the FTIR signals of the resins may overlap with those of the OM in the sample (Arocena et al., 1995). Furthermore, the resin may interact with mineral soil components (i.e., C=O functional groups with Ca^{2+} or Fe^{3+} ions), causing shifts in absorption bands that complicate the interpretation of the spectra (Dupuy and Douay, 2001; Aparicio et al., 2002). The standard DRIFT technique is based on finely ground and highly diluted samples (Hesse et al., 1984; Tremblay and Gagné, 2002), but for analyzing SOM content, the dilution may not always be required (e.g., Reeves, 2003). For relatively smooth surfaces, the DRIFT technique could be applied to analyze differences in OM composition along transects at intact aggregate surfaces (Ellerbrock et al., 2009).

The small-scale spatial distribution in the physical and chemical properties of flow-path surfaces may amplify preferential flow effects during the transport of reactive solutes (e.g., Flühler et al., 1996; Bundt et al., 1999, 2000). Small-scale heterogeneities were proven for soil chemical properties of different aggregate fractions (Wilcke and Kaupenjohann, 1994) and increased clay (Horn, 1987) and C contents of aggregate coatings in comparison to the cores and the soil matrix (Kaiser and Wilcke, 1996; Sullivan and Koppi, 1991; Park and Smucker, 2005). The OM may have also been transported into preferential flow paths, as indicated by ^{14}C radiocarbon dating on OM from earthworm burrows (Don et al., 2008) and from refilled macropores (Kayser et al., 1994; Chabbi et al., 2009).

Small-scale heterogeneities in SOM distribution at flow-path surfaces seem to be important, especially for pesticide transport and chemical sorption processes in structured soils (e.g. White, 1985; Jury and Flühler, 1992; Kaiser and Zech, 1998). Local differences in wettability were observed (Hallett et al., 2004), which may cause spatial differences in water movement (Lipsius and Mooney, 2006). While aggregate biopore surfaces are intensively studied with respect to their mineralogy (e.g., Kaiser and Wilcke, 1996; Celis et al., 1997; Jongmans et al., 1998), only a few studies (Ellerbrock and Gerke, 2004; Ellerbrock et al., 2009; Guggenberger et al., 1996) have focused on SOM composition at such internal soil structural surfaces in the context of preferential flow. The three-dimensional spatial distribution of local OM properties along such aggregate and biopore surfaces with thin and vulnerable coatings is largely unknown.

Our objective was to use a DRIFT mapping procedure to analyze the local distribution of OM composition at the intact surfaces of different aggregates and worm burrows that could potentially act as preferential flow paths. For the DRIFT analyses, we used intact samples from aggregated soils where preferential flow was previously observed (Kodesova et al., 2008). A comparison between OM composition and wettability on the basis of droplet infiltration tests will be discussed as a preliminary interpretation of the mapping results. The novelty of this research is the analysis of small-scale distribution of OM composition at intact surfaces in structured soil.

Materials and Methods

Sample Preparation

Soil samples were taken from the Bt horizons (60–75 cm depth) of two loess-derived Alfisol profiles (Soil Survey Staff, 2006) located at adjacent grassland (G) and arable (Ar) sites at Hněvčevy (near Hradec Králové, Czech Republic). For more detailed information on the Hněvčevy site and the soils see Kodesova et al. (2008, 2009).

Soil blocks of approximately 20- by 30- by 10-cm edge lengths were cut or broken out of the soil pits using a spade and packed in

aluminum foil to prevent possible evaporation or cracking. From these blocks, smaller subsamples or clods of about 5- to 10-cm edge length were manually separated along existing surface cracks and air dried to obtain relatively intact crack surfaces. Sections of 3 by 4 cm up to 6 by 8 cm were separated by scraping material from the surrounding clod and from below (i.e., to reduce the sample thickness to a maximum of 15 mm) with a knife. These samples were wrapped in aluminum foil to stabilize the sides (Ellerbrock et al., 2009) and dried over silica gel in a dessicator. The regions with earthworm burrows, root channels, and aggregate coatings at the surface areas were identified by visual inspection.

We selected one intact surface area sample with two earthworm burrows from the G profile and three intact crack surfaces that show partial root channels and clay-organic coatings from the Ar profile. Polystyrene beads (2-mm diam.) in bores at the corners and polyethylene stripes at the edges (G) of the samples—their spectral signals at 3000 and 2800 cm^{-1} are significantly different from those of the soil (Gottwald and Wachter, 1997)—served as markers for later spatial allocation of the spectroscopic data. The stabilized aggregate samples were fixed onto aluminum plates that were equipped with a millimeter scale to facilitate the spatial allocation of the DRIFT data.

DRIFT Mapping of Intact Surfaces

The DRIFT mapping is defined here as a collection of two-dimensional distributed DRIFT data from numerous measurements at a sample surface obtained by combining a DRIFT device with an *XY*-positioning table. While previously (Ellerbrock et al., 2009) a stepwise manual movement of a 10- by 60-mm sample was performed, the *XY*-positioning device (Resultec, Illerkirchberg, Germany) allowed automatic acquisition of DRIFT spectra from a maximal area of 8 by 12 cm. Automatic DRIFT mapping was performed for several aggregate surfaces in 1.0-mm steps along parallel transects of 1.0-mm distance. After each individual measurement, the sample was moved forward in 1.0-mm steps before starting the next scan. Note that the IR beam covers a circular area of about 1.5 mm in diameter. For each spectrum, 16 scans were compiled to give one spectrum between 4000 and 400 cm^{-1} at a resolution of 4 cm^{-1} . A gold target (99% Infragold, Labsphere, North Sutton, NH) for the background spectra measurement was fixed onto the positioning table in the averaged height of the sample surfaces.

Spectra Processing and Analyses

For spectra processing, the software WIN-IR Pro 3.4 (Digilab, Holliston, MA) was used. The reflectance data measured for the crack surfaces were converted to Kubelka–Munk (KM) units (Kubelka, 1948, based on Kubelka and Munk, 1931; Ciani et al., 2005) by using the standard KM transformation as implemented in the software. The spectra were smoothed with the boxcar moving average algorithm applying a factor of 25. Baseline correction was performed from 4000 cm^{-1} and a second point in the range from 1200 to 1000 cm^{-1} . The obtained DRIFT spectra were analyzed

semiquantitatively in Region A (3000–2800 cm^{-1}) and Region B (1740–1698 cm^{-1} and 1640–1600 cm^{-1}).

The absorption intensities of C–H stretch vibrations of alkyl groups within the range between 3020 and 2800 cm^{-1} are usually superimposed as a shoulder of the broad OH band reaching from 3750 to 2700 cm^{-1} (Hesse et al., 1984). Therefore the intensities of bands A were measured as the vertical distance from a local “baseline” plotted between tangential points on absorption minima in the region to each of the maxima (Capriel et al., 1995; Ellerbrock et al., 2009). The peak centers were located between 2948 and 2920 cm^{-1} for asymmetric C–H (A_a) stretch vibrations and between 2864 and 2849 cm^{-1} for symmetric (A_s) stretch vibrations. Both height values of A_a and A_s bands were added and denoted as Band A. The heights from the baseline of the absorption bands at 1710 cm^{-1} and 1620 cm^{-1} were added and denoted as Band B. The absorption band at 1710 cm^{-1} refers to groups with C=O double bonds (Celi et al., 1997) that are part of ketones, carboxylic acids, or amides (MacCarthy and Rice, 1985; Hesse et al., 1984). The absorption band at 1620 cm^{-1} refers to groups with C=O double bonds that are part of carboxylate anions and to aromatic groups if these are conjugated with C=O groups such as unsaturated ketones (Gottwald and Wachter, 1997). The bands at 1660 to 1600 cm^{-1} may also refer to the OH bending vibrations of water molecules in hydration layers of soil phyllosilicates (Mendelovici et al., 1995; Spaccini et al., 2001). However, it is assumed that absorption band B reflects mainly the content of hydrophilic groups in soil since (i) C=O containing groups are hydrophilic; (ii) aromatic rings, although generally assumed to indicate hydrophobic groups, can show hydrophilic properties if they are conjugated with C=O groups (Morrison and Boyd, 1983); and (iii) water molecules in hydration layers also indicate functional groups with hydrophilic behavior. Thus, for our purposes, the bands at 1740 to 1698 and 1640 to 1600 cm^{-1} (i.e., those that are indicating functional groups that behave similarly hydrophilic) were combined as absorption band B. The cumulative peak heights of bands A were related to those of the bands B. The ratio of hydrophobic (i.e., aliphatic, region A in FTIR spectra) to hydrophilic (i.e., carbonyls, etc., region B in FTIR spectra) functional groups (Fig. 1b) may be used as measure of the potential wettability of the soil OM (Ellerbrock et al., 2005).

Droplet Infiltration Tests

First droplet infiltration tests were performed at crack surface CZ-1 to see whether the spatial patterns of A/B ratios possibly correspond with local differences in water absorption properties or indirectly with the wettability along the flow-path surfaces. Water drop infiltration times were measured by using a contact angle device that was coupled with a high-speed camera (OCA 20, DataPhysics, Filderstadt, Germany). A droplet of 4.9 μL of deionized water from a 1-mL syringe (Hamilton, Reno, NV) was placed on visually selected surface areas of (i) the uncoated crack surface, (ii) the older burrow, and (iii) the younger earthworm burrow. The penetration of the water drop was

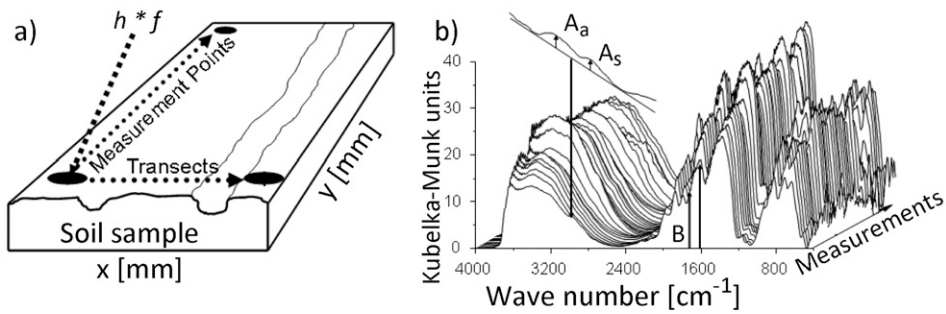


Fig. 1. Illustration of the basic principle of DRIFT mapping: (a) generalized intact soil sample on which DRIFT measurements were performed at single measurement points (y direction, 1-mm steps) along parallel transects (x direction, 1-mm distance), $h * f$ = IR beam; (b) Kubelka-Munk (KM) transformed spectra derived from DRIFT data along a transect, showing the analyzed absorption bands A (A_a , asymmetric, and A_s , symmetric, C-H stretch) and bands B (C=O stretch).

recorded at a rate of 350 Hz and analyzed by the software SCA (OCA 20, DataPhysics) for the time required for the water drop to completely infiltrate into the sample surface. The relatively rough surface of aggregate and burrows did not allow measurement of contact angles.

Results

The crack surface of sample CZ-1 (Fig. 2) from the grassland soil had two earthworm burrows arranged as parallel channels with a diameter of about 5 mm (left) and 7 mm (right) within a relatively smooth crack surface. The burrows were vertically oriented in the soil. From visual inspection, the left burrow was assumed to be older because it contained numerous pores and some filled-in material. In contrast, the right-hand side burrow was classified as younger (i.e., more recently created). One single root was detected in the lower part of the sample (Fig. 2a).

DRIFT mapping of the field moist crack surface CZ-1 (Fig. 2) resulted in a distributed map of A/B ratios from the KM-transformed spectra ranging between 0 and 0.033. Most

crack surface of the “smooth” plateau area had A/B ratios between 0 and 0.009, whereas the A/B ratios, especially of the younger earthworm burrow (max. A/B ratios of 0.033), were up to 10 times higher (Fig. 2b). The older earthworm burrow had only slightly elevated values of the A/B ratios (i.e., max. 0.018) as compared with the plateau. The A/B ratios of the area with the individual root corresponded to those of the older burrow. Polystyrene beads were used as artificial markers showing the highest A/B ratios (max. 0.277). For better clarity, the ratios on the color scale were limited to a value of 0.033.

DRIFT mapping of the air-dried crack surface CZ-1 (Fig. 2c) exhibited A/B ratios of the plateau crack surface region mainly between 0 and 0.009; the proportion of low-valued A/B ratios (0–0.006) was larger than for the moist sample. The older earthworm burrow could be identified more distinctively from the contrasting plateau area, although A/B ratios remained at the same level as for the moist sample. The A/B ratios of the root area were slightly higher (max. 0.021) as compared with the moist sample. The ratios for most of the younger burrow areas increased up to the maximal value of 0.033 for A/B; at some points not separately shown, the A/B values were even higher than 0.033.

The DRIFT mapping of the dried sample surface in 1-mm steps perpendicular to the orientation of the earthworm burrows (Fig. 2, arrow) showed the entire range of A/B values of the preferential flow-path surface CZ-1 (Fig. 3). The plateau of the uncoated crack surface (see ii) exhibited relatively low values of A/B as compared to the earthworm burrows, especially the younger burrow (see

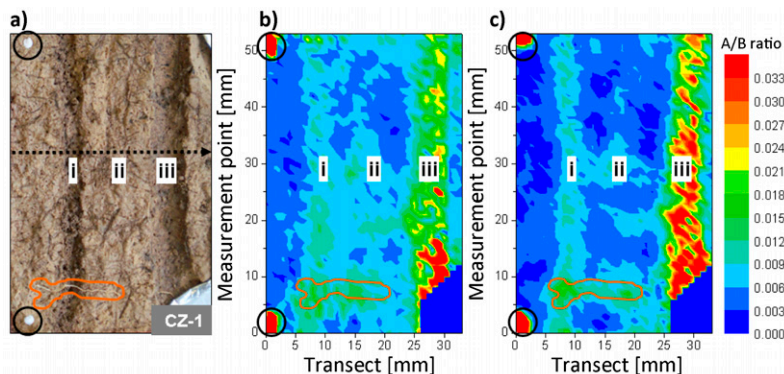


Fig. 2. (a) Photo and map of A/B ratios of (b) a “field-moist,” and (c) a desiccator-dried crack surface (CZ-1: Hnevceves, G, Luvisol, Bt horizon, 60–80 cm depth). The white marks indicate an older earthworm burrow (i), the crack surface plateau (ii) (i.e., mostly uncoated matrix of crack surface), and the younger worm burrow (iii). The orange-framed region indicates a root. Black circles denote polystyrene markers. The dashed arrow indicates the measurement points at position 37 across the transects for which the A/B ratios are shown in Fig. 3.

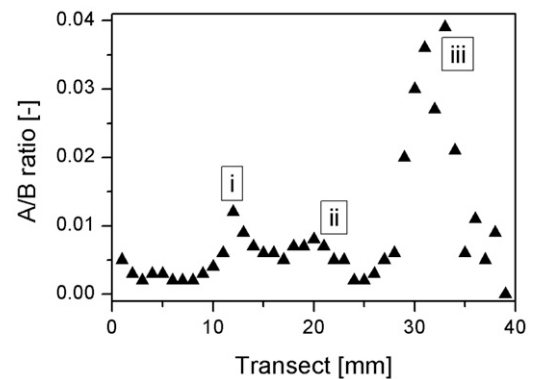


Fig. 3. A/B ratios obtained from DRIFT spectra of an air-dried aggregate surface (CZ-1: Hnevceves, G, Luvisol, Bt horizon, 60–80 cm depth). Data are from measuring points at position 37 at the transects 1 through 38, from left to right (perpendicular to the vertical orientation). Note older earthworm burrow (i), plateau (ii), uncoated crack surface, and younger burrow (iii).

iii). The region of the older earthworm burrow at crack surface CZ-1 shows a mean drop infiltration time (245 ms; min. 128 ms; max. 400 ms), which was significantly lower than that obtained from the uncoated crack surface (798 ms; min. 353 ms; max. 1105 ms) or from the surface of the younger earthworm burrow (1355 ms; min. 370 ms; max. 2263 ms).

The crack surface of sample CZ-9 was relatively smooth (Fig. 4a) and contained shallow brownish indentations of fine roots. According to visual inspection, small areas were coated with clay-organic material. In other areas, fine roots were observed. The A/B ratios determined for the crack surface were mainly between 0.003 and 0.009 (Fig. 4b). However, in the region of a 1-cm-long root channel (transect points 36 to 45) the A/B ratios were up to 0.024. Higher A/B ratios were not found for CZ-9 except for the polystyrene markers.

The crack surface CZ-10 showed a slightly rough and irregular sloped surface with root channels having brown colored coatings visible to the naked eye (Fig. 4c). The A/B ratios of about 0.003 to 0.006 for most of the CZ-10 surface area were relatively homogeneously distributed. Only the coated region with a larger number of root channels in the center of the sample showed higher A/B ratios up to values of 0.021 (Fig. 4d). Note that the root channel of approximately 1-cm length located in the upper right area of the sample was not reflected in the map of A/B ratios.

The crack surface CZ6 was relatively smooth but sloped, as is shown by the blurred upper half of the sample area (Fig. 5a). Most of the surface exhibited low A/B values (0–0.009). However, the upper border of the sample displayed higher A/B up to 0.024 (Fig. 5b). Note that the root residue at the even surface (lower part of the sample) area corresponded to higher A/B (max. 0.021) than the root residue of the upper sloped area (max. 0.009).

Discussion

DRIFT mapping indicates a small-scale heterogeneity of both OM composition and distribution at intact surfaces of preferential flow paths at the arable (Ar) (Fig. 4 and 5) and the grassland (G) (Fig. 2) samples. The spatial distribution of the A/B values fits well to the visual classification of the flow-path surfaces (matrix, clay-organic coatings, earthworm burrows, root channels, roots). Uncoated crack surface areas (as it is visually assessable) show lowest A/B ratios of 0 to 0.009 that were found on the surfaces reflecting a relatively high potential wettability of the flow path surface (Hallett et al., 2004). The A/B ratios in the range of zero can be explained by the absence of organic material or by absence of aliphatic groups in

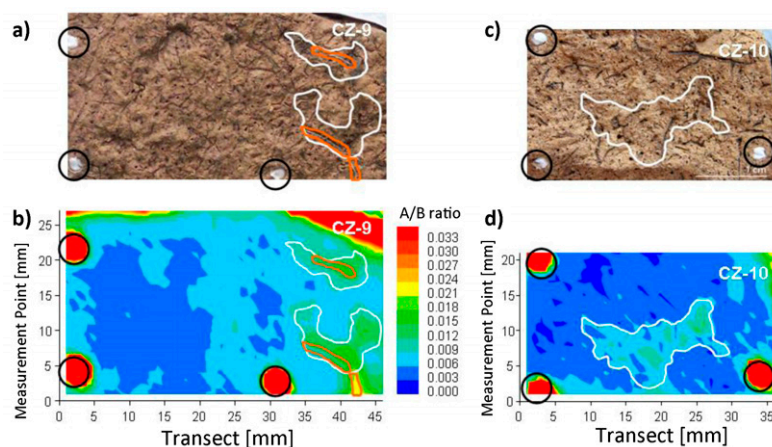


Fig. 4. (a, c) Photos and (b, d) A/B ratios of air-dried crack surfaces CZ-9 and CZ-10, respectively, with partly coated biopores (Hneveves, arable land, Luvisol, Bt horizon, 60–75 cm depth). White-framed areas show clay organic matter coatings (CZ-10: coatings in root channels); orange-framed areas show roots. Black circles denote polyethylene markers.

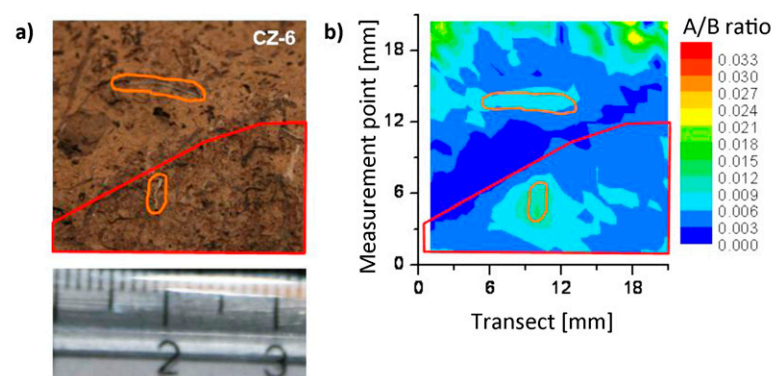


Fig. 5. (a) Photo and (b) A/B ratios of an air-dried and uneven crack surface (CZ6: Hneveves, arable land, Luvisol, Bt horizon, 60–75 cm depth). Red-framed area shows even surface area, with the upper part of the sample sloped. Orange-framed area shows roots.

the analyzed material. In contrast, high A/B ratios (0.021–0.033) occurring at coated surface areas (clay-organic material or earthworm slime) are supposed to have a lower potential wettability.

The A/B ratios of the uncoated crack surface of the Luvisols studied here are three to five times larger than the A/B of an Alfisol at an arable site (0.001–0.002) described by Ellerbrock et al. (2009). The A/B ratios in the region the earthworm burrows are about three to six times (older burrow) up to 15 times (younger burrow) larger than earlier findings (Ellerbrock et al., 2009) for an earthworm burrow of the Alfisol (i.e., with A/B values of 0.0015–0.0025).

For comparing DRIFT mapping results of different samples the consideration of the texture at the samples surface is of great importance since the particle size affects the signal intensity to a great extent (Kortüm, 1969; Ventura and Papini, 1997). Note that all results were shown in KM units, although KM transformation

is not appropriate in a strict sense since samples are neither diluted nor smooth (Gottwald and Wachter, 1997). Preliminary investigations indicated that KM transformation of DRIFT signals may be permitted as far as the particle size remains constant and $<70\ \mu\text{m}$ (data not published).

Differences in terms A/B between the grassland and the arable samples did not appear in the case of the uncoated or thin coated matrix. However, small-scaled surface property changes occur at earthworm burrows (Fig. 2) and clay-organic coatings in connection with rooting of the flow paths (Fig. 4a and 4b). The A/B ratios detected for the younger earthworm burrow at CZ-1 (0.024–0.033) are 10 to 12 times larger than for the uncoated matrix according to findings of Guggenberger et al. (1996), who used ^{13}C nuclear magnetic resonance (cross polarization magic angle spinning) spectral data to observe a higher alkyl C/carboxyl C ratio (A/B ratio) for the casts as compared with the surrounding matrix. DRIFT data indicate that for the younger burrow the relation of hydrophobic (alkyl; Band A) to hydrophilic (carboxyl; Band B) components is higher compared to the older burrow, showing the degradation of the OM at the layers of earthworm burrows (Schrader and Zhang, 1997; Don et al., 2008).

The DRIFT mapping procedure was found suitable to detect small-scale differences on surfaces of preferential flow paths for features larger than 0.5 mm. Features such as fine root channels (Fig. 4c and 4d) cannot be distinguished by DRIFT mapping. The size limitation is attributed to the relation between measurement device geometry and both the size and the spectral properties of the feature (Albertz, 2007). In the case of CZ-10 (Fig. 4c, upper right corner of the sample) the fine root channel has a diameter of less than 0.5 mm in comparison to 1.5 mm for the infrared beam. Additionally, the differences between the spectral properties of root channel and matrix are too small to produce a high spectral contrast.

The results of CZ-6 (Fig. 5) illustrate additional problems of DRIFT mapping of undisturbed (in situ) soil surfaces. First, both the slope and elevation of the sample surface can lead to reduced DRIFT signal intensity if the measurement points are located outside the focus of the parabolic mirror of the device. For example, for the root in the upper part of sample CZ-6 (Fig. 5), the A/B values are not as high as for the root in the lower part of the sample, even though both roots were expected to be identical. It is possible that the reduced signal intensity primarily affects the small A bands (C–H bonds). Second, the sample surface topology affects the geometric position of the measurement point by causing an offset of the IR beam if the point is out of focus. For example, if the orientation of the root in the upper part is perpendicular to that in the lower part of sample CZ-6 (Fig. 5a), this is reflected by the A/B values in the marked surrounding area (Fig. 5b). Such problems are well known in remote sensing research, and radiometric correction of the spectral data has been proposed (Stohr and West, 1975; Wu et al., 2008; Mishra et al., 2009).

It is commonly known that high water contents of the sample influences the DRIFT signal intensities in the range from 3800 to 2400 cm^{-1} (e.g., Gottwald and Wachter, 1997). In the case of a moist sample, the signal intensities of the broad O–H band at about $\text{WN } 3400\ \text{cm}^{-1}$ overlie the relatively weak C–H bands such that bands A are more difficult to detect than in a sample with lower water content. To minimize variable moisture content effects, the samples were dried in a silica-filled dessicator up to mass constancy to obtain reproducible results. The maps of A/B ratios for field-moist and dessicator dry samples are shown in Fig. 2.

Assuming that the A/B ratios reflect the potential wettability of the samples (Ellerbrock et al., 2005), the water drop infiltration times appear to reflect differences in A/B ratios. As assumed, the mean drop infiltration time for the region of older earthworm burrow (245 ms) at the crack surface CZ-1 was significantly lower than for the surface of the younger earthworm burrow (1355 ms). In contrast to the assumed relation between A/B and wettability, the infiltration times were relatively shorter for the region of the older earthworm burrow than for those of the uncoated crack surface (798 ms), despite relatively higher A/B ratios for the burrow wall. Of course, water infiltration also depends on porosity and other properties of the surface layer in addition to the A/B ratio (i.e., the older burrow wall has a higher number of pores compared with the younger one; Fig. 2a). Furthermore, we note that the water drop covers an area of about 19.6 mm^2 , while the A/B ratios determined from DRIFT mapping reflect the properties of only 1.2 mm^2 . To link the A/B ratios with water infiltration, additional studies are needed, including a spatial allocation of drop and DRIFT measurement positions as well as a spatial interpolation of A/B ratios across the droplet-affected area. Further analyses, however, are beyond the scope of this discussion.

Conclusions

The results of this research allow the following conclusions with respect to the small-scale distribution of OM-related surface properties of preferential flow paths in structured soil. With respect to methodology, the DRIFT mapping technique is useful for detecting two-dimensional millimeter-scale distributions of the OM composition at intact sample surfaces. The spatial differences in CH functional groups on internal structures (i.e., coated crack and biopore surfaces) can be better represented for dry than for moist conditions. For a fine-textured (loess) soil, three types of structural surfaces of preferential flow paths were distinguished by mapping the CH and CO functional groups in OM in terms of the A/B ratios. For walls of earthworm burrows and root channels, and clay-organic coatings, the A/B ratios were higher than those of the surrounding surfaces of the uncoated porous soil matrix. Sloped sample surfaces showed reduced reflectance values, indicating that the surface topography needs to be considered when interpreting DRIFT signal intensities. Comparative water drop infiltration tests suggest that it is possible that hydraulic properties

correspond to the spatial distribution of A/B ratios, indicating local differences in potential wettability. The locally distributed OM composition of coatings at preferential flow-path surfaces indicates yet unknown implications for preferential movement of water and, especially, of reactive solutes.

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References

- Albertz, J. 2007. Einführung in die Fernerkundung. 3. (In German.) Aufl., Wiss. Buchgesellschaft Darmstadt, Germany.
- Aparicio, S., S.B. Doty, N.P. Camacho, E.P. Paschalis, L. Spevak, R. Mendelsohn, and A.L. Boskey. 2002. Optimal methods for processing mineralized tissues for Fourier Transform infrared microspectroscopy. *Calif. Tissue Int.* 70:422–429.
- Arocena, J.M., S. Pawluk, M.J. Dudas, and A. Gajdostik. 1995. In situ investigation of soil organic matter aggregates using infrared microscopy. *Can. J. Soil Sci.* 75:327–332.
- Beven, K., and P. Germann. 1982. Macropores and water flow in soils. *Water Resour. Res.* 18:1311–1325.
- Bundt, M., A. Albrecht, P. Froidevaux, P. Blaser, and H. Flüher. 2000. Impact of preferential flow on radionuclide distribution in soil. *Environ. Sci. Technol.* 34:3895–3899.
- Bundt, M., F. Widmer, P. Blaser, and H. Flüher. 1999. Präferentieller Wasserfluss—Auswirkungen eines bodenphysikalischen Phänomens auf chemische und biologische Parameter. (In German, with English abstract.) *Mitteilg. Dtsch. Bodenkundl. Gesellsch.* 91:314–317.
- Capriel, P. 1997. Hydrophobicity of organic matter in arable soils: Influence of management. *Eur. J. Soil Sci.* 48:457–462.
- Capriel, P., T. Beck, H. Borchert, J. Gronholz, and G. Zachmann. 1995. Hydrophobicity of the organic matter in arable soils. *Soil Biol. Biochem.* 27:1453–1458.
- Celi, L., M. Schnitzer, and M. Nègre. 1997. Analysis of carboxyl groups in soil humic acids by a wet chemical method, Fourier-transform infrared spectroscopy and solution-state carbon-13 nuclear magnetic resonance: A comparative study. *Soil Sci.* 162:189–197.
- Celis, R., L. Cox, M.C. Hermosin, and J. Cornejo. 1997. Sorption of thiazafurion by iron- and humic acid-coated montmorillonite. *J. Environ. Qual.* 26:472–479.
- Chabbi, A., I. Kögel-Knabner, and C. Rumpel. 2009. Stabilised carbon in subsoil horizons is located in spatially distinct parts of the soil profile. *Soil Biol. Biochem.* 41:256–261.
- Ciani, A., K.-U. Goss, and R.P. Schwarzenbach. 2005. Light penetration in soil and particulate materials. *Eur. J. Soil Sci.* 56:561–574.
- Don, A., B. Steinberg, I. Schöning, K. Pritsch, M. Joschko, G. Gleixner, and E.-D. Schulze. 2008. Organic carbon sequestration in earthworm burrows. *Soil Biol. Biochem.* 40:1803–1812.
- Dupuy, N., and F. Douay. 2001. Infrared and chemometrics study of the interaction between heavy metals and organic matter in soil. *Spectrochim. Acta [A]* 57:1037–1047.
- Ellerbrock, R.H., and H.H. Gerke. 2004. Characterizing organic matter of soil aggregate coatings and biopores by Fourier transform infrared spectroscopy. *Eur. J. Soil Sci.* 55:219–228.
- Ellerbrock, R.H., H.H. Gerke, J. Bachmann, and M.-O. Goebel. 2005. Composition of organic matter fractions for explaining wettability of three forest soils. *Soil Sci. Soc. Am. J.* 69:57–66.
- Ellerbrock, R.H., H.H. Gerke, and C. Böhm. 2009. In situ DRIFT characterization of organic matter compositions on soil structural surfaces. *Soil Sci. Soc. Am. J.* 73:531–540.
- Flüher, H., W. Durner, and M. Flury. 1996. Lateral solute mixing processes—A key for understanding field-scale transport of water and solutes. *Geoderma* 70:165–183.
- Flury, M., H. Flüher, W.A. Jury, and J. Leuenberger. 1994. Susceptibility of soils to preferential flow of water: A field study. *Water Resour. Res.* 30:1945–1954.
- Gerzabek, M.H., F. Pichlmayer, H. Kirchmann, and G. Haberhauer. 1997. The response of soil organic matter to manure amendment in a long-term experiment at Ultuna, Sweden. *Eur. J. Soil Sci.* 48:273–282.
- Ghodrati, M., and W.A. Jury. 1992. A field study of the effects of soil structure and irrigation method on preferential flow of pesticides in unsaturated soil. *J. Contam. Hydrol.* 11:101–125.
- Gottwald, W., and G. Wachter. 1997. IR spectroscopy for practitioners. (In German: IR Spektroskopie für Anwender.) Wiley-VCH, Weinheim, Germany.
- Gregoriou, V.G., and S.E. Rodman. 2002. Quantitative depth profile analysis of micrometer-thick multilayered thin coatings using step-scan FT-IR photoacoustic spectroscopy. *Anal. Chem.* 74:2361–2369.
- Guggenberger, G., R.J. Thomas, and W. Zech. 1996. Soil organic matter within earthworm casts of an anecic-endogeic tropical pasture community, Colombia. *Appl. Soil Ecol.* 3:263–274.
- Haberhauer, G., and M.H. Gerzabek. 1999. Drift and transmission FT-IR spectroscopy of forest soils: An approach to determine decomposition processes of forest litter. *Vib. Spectrosc.* 19:413–417.
- Hallett, P.D., N. Nunan, J.T. Douglas, and I.M. Young. 2004. Millimeter-scale spatial variability in soil water sorptivity: Scale, surface elevation, and subcritical repellency effects. *Soil Sci. Soc. Am. J.* 68:352–358.
- Hayes, M.H.B., and C.E. Clapp. 2001. Humic substances: Considerations of compositions, aspects of structure, and environmental influences. *Soil Sci.* 166:723–737.
- He, W., R. Shanks, and G. Amarasinghe. 2002. Analysis of additives in polymers by thin-layer chromatography coupled with Fourier transform-infrared spectroscopy. *Vib. Spectrosc.* 30:147–156.
- Hesse, M., H. Meier, and B. Zeeh. 1984. Spectroscopic methods in organic chemistry. (In German: Spektroskopische Methoden in der organischen Chemie.) Georg Thieme, Stuttgart, Germany.
- Horn, R. 1987. Die Bedeutung der Aggregation für die Nährstoffsorption in Böden. (In German, with English abstract.) *J. Plant Nutr. Soil Sci.* 150:13–16.
- Jarvis, N.J. 2007. A review of non-equilibrium water flow and solute transport in soil macropores: Principles, controlling factors and consequences for water quality. *Eur. J. Soil Sci.* 58:523–546.
- Jongmans, A.G., L. Denaix, and F. van Oort. 1998. Isotropic smectite coatings and spheroids in holocene volcanic deposits under temperate climate, France. *Soil Sci. Soc. Am. J.* 62:287–295.
- Jury, W.A., and H. Flüher. 1992. Transport of chemicals through soil: Mechanisms, models, and field application. *Adv. Agron.* 47:141–201.
- Kaiser, M., R.H. Ellerbrock, and H.H. Gerke. 2007. Long-term effects of crop rotation and fertilization on soil organic matter composition. *Eur. J. Soil Sci.* 58:1460–1470.
- Kaiser, M., R.H. Ellerbrock, and H.H. Gerke. 2008. Cation exchange capacity and composition of soluble soil organic matter fractions. *Soil Sci. Soc. Am. J.* 72:1278–1285.
- Kaiser, K., K. Eusterhues, C. Rumpel, G. Guggenberger, and I. Kögel-Knabner. 2002. Stabilization of organic matter by soil minerals—investigations of density and particle-size fractions from two acid forest soils. *J. Plant Nutr. Soil Sci.* 165:451–459.
- Kaiser, K., and W. Wilcke. 1996. Pedogenetic differentiation of soil properties in aggregates. (In German, with English abstract.) *J. Plant Nutr. Soil Sci.* 159:599–603.
- Kaiser, K., and W. Zech. 1998. Soil dissolved organic matter sorption as influenced by organic and sesquioxide coatings and sorbed sulfate. *Soil Sci. Soc. Am. J.* 62:129–136.
- Kayser, A., W. Wilcke, M. Kaupenjohann, and J.D. Joslin. 1994. Small scale heterogeneity of soil chemical properties. I. A technique for rapid aggregate fractionation. *J. Plant Nutr. Soil Sci.* 157:453–458.
- Kögel-Knabner, I. 1997. C-13 and N-15 NMR spectroscopy as a tool in soil organic matter studies. *Geoderma* 80:243–270.
- Köhne, J.M., S. Köhne, and J. Simunek. 2009a. A review of model applications for structured soils: a) Water flow and tracer transport. *J. Contam. Hydrol.* 104:4–35.
- Köhne, J.M., S. Köhne, and J. Simunek. 2009b. A review of model applications for structured soils: b) Pesticide Transport. *J. Contam. Hydrol.* 104:36–60.
- Kodesova, R., M. Kocarek, V. Kodes, J. Simunek, and J. Kozak. 2008. Impact of soil micromorphological features on water flow and herbicide transport in soils. *Vadose Zone J.* 7:798–809.
- Kodesova, R., N. Vignozzi, M. Rohoskova, T. Hajkova, M. Kocarek, M. Pagliai, and J. Simunek. 2009. Impact of varying soil structure on transport processes in different diagnostic horizons of three soil types. *J. Contam. Hydrol.* 104:107–125.
- Kortüm, G. 1969. Reflexionsspektroskopie. (In German.) Springer-Verlag, Berlin.
- Kubelka, P. 1948. New contributions to the optics of intensely light-scattering materials. Part I. *J. Opt. Soc. Am.* 38:448–457.
- Kubelka, P., and F. Munk. 1931. Ein Beitrag zur Optik der Farbanstriche. (In German.) *Z. Tech. Phys.* 11:593–601.
- Lipsius, K., and S.J. Mooney. 2006. Using image analysis of tracer staining to examine the infiltration patterns in a water repellent contaminated sandy soil. *Geoderma* 136:865–875.

- Ludwig, B., B. John, R.E. Ellerbrock, M. Kaiser, and H. Flessa. 2003. Stabilization of carbon from maize in a sandy soil in a long-term experiment. *Eur. J. Soil Sci.* 54:117–126.
- MacCarthy, P., and J.A. Rice. 1985. Spectroscopic methods (other than NMR) for determining functions in humic substances. p. 527–560. *In* G.R. Aiken et al. (ed.) *Humic substances in soil, sediment, and water: Geochemistry, isolation, and characterization*. Wiley Interscience, New York.
- Ma'shum, M., and V.C. Farmer. 1985. Origin and assessment of water repellency of a sandy South Australian soil. *Aust. J. Soil Res.* 23:623–626.
- Mendelovici, E., R. Villalba, A. Sagarzazu, and O. Carias. 1995. The 1640 cm^{-1} infrared band, monitor for the gain and thermal stability of water produced in ground kaolinites. *Clay Miner.* 30:307–313.
- Mertig, M., D. Klemm, H. Zänker, and W. Pompe. 2002. Scanning force microscopy of two dimensional structure formation in thin humic acid films. *Surf. Interface Anal.* 33:113–117.
- Mishra, V.D., J.K. Sharma, K.K. Singh, N.K. Thakur, and M. Kumar. 2009. Assessment of different topographic corrections in AWiFS satellite imagery of Himalaya terrain. *J. Earth Syst. Sci.* 118:11–26.
- Morrison, R.T., and R.N. Boyd. 1983. *Lehrbuch der organischen Chemie*. (In German.) Verlag Chemie, Weinheim, Germany.
- Negre, M., D. Vindrola, S. Spera, L. Ferraris, and M. Gennai. 2002. Effect of the chemical composition of soil humic acids on their viscosity, surface pressure, and morphology. *Soil Sci.* 167:636–651.
- Nieber, J. 2000. The relation of preferential flow to water quality, and its theoretical and experimental quantification. p. 1–10. *In* D. Bosch et al. (ed.) *Preferential flow, water movement and chemical transport in the environment*. Proc. 2nd Int. Symp., Honolulu, HI. 3–5 Jan. 2001. ASAE, St. Joseph, MI.
- Park, E.J., and A.J.M. Smucker. 2005. Erosive strengths of concentric regions within soil macroaggregates. *Soil Sci. Soc. Am. J.* 69:1912–1921.
- Reeves, J.B. 2003. Mid-infrared diffuse reflectance spectroscopy: Is sample dilution with KBr necessary, and if so, when? *Am. Lab.* 35:24–25.
- Reeves, J.B., B.A. Francis, and S.K. Hamilton. 2005. Specular reflection and diffuse reflectance spectroscopy of soils. *Appl. Spectrosc.* 59:39–46.
- Reeves, J.B., G.W. McCarty, and V.B. Reeves. 2001. Mid-infrared diffuse reflectance spectroscopy for the quantitative analysis of agricultural soils. *J. Agric. Food Chem.* 49:766–772.
- Schrader, S., and H.Q. Zhang. 1997. Earthworm casting: Stabilization or destabilization of soil structure? *Soil Biol. Biochem.* 29:469–475.
- Soil Survey Staff. 2006. *Keys to soil taxonomy*. 10th ed. USDA-NRCS, Washington, DC.
- Spaccini, R., A. Piccolo, G. Haberhauer, M. Stemmer, and M.H. Gerzabek. 2001. Decomposition of maize straw in three European soils as revealed by DRIFT spectra of soil particle fractions. *Geoderma* 99:245–260.
- Sposito, G. 1989. *The chemistry of soils*. Oxford Univ. Press, New York.
- Stohr, C.J., and T.R. West. 1975. Terrain and look angle effects upon multi-spectral scanner response. *Photogramm. Eng. Remote Sens.* 51:229–235.
- Sullivan, L.A., and A.J. Koppy. 1991. Morphology and genesis of silt and clay coatings in the vesicular layer of a desert loam soil. *Aust. J. Soil Res.* 29:579–586.
- Tatzber, M., M. Stemmer, H. Spiegel, C. Katzlberger, G. Haberhauer, A. Mentler, and M.H. Gerzabek. 2007. FTIR-spectroscopic characterization of humic acids and humin fractions obtained by advanced NaOH , $\text{Na}_2\text{P}_2\text{O}_7$, and Na_2CO_3 extraction procedures. *J. Plant Nutr. Soil Sci.* 170:522–529.
- Tremblay, L., and J.-P. Gagné. 2002. Fast quantification of humic substances and organic matter by direct analysis of sediments using DRIFT spectroscopy. *Anal. Chem.* 74:2985–2993.
- Urbanek, E., P. Hallett, D. Feeney, and R. Horn. 2007. Water repellency and distribution of hydrophilic and hydrophobic compounds in soil aggregates from different tillage systems. *Geoderma* 140:147–155.
- Van der Marel, H.W., and H. Beutelspacher. 1976. *Atlas of infrared spectroscopy of clay minerals and their admixtures*. Elsevier, Amsterdam.
- Ventura, C., and M. Papini. 1997. Study of the spectral properties of inhomogeneous granular materials. *Macromol. Symp.* 119:137–148.
- von Lützw, M., I. Kögel-Knabner, K. Ekschmitt, H. Flessa, G. Guggenberger, E. Matzner, and B. Marschner. 2007. SOM fractionation methods: Relevance to functional pools and to stabilization mechanisms. *Soil Biol. Biochem.* 39:2183–2207.
- White, R.E. 1985. The analysis of solute breakthrough curves to predict water redistribution during unsteady-flow through undisturbed structured clay soil. *J. Hydrol.* 79:21–35.
- Wilcke, W., and M. Kaupenjohann. 1994. Small scale heterogeneity of soil chemical properties. II. Fractions of aluminum and heavy metals. *J. Plant Nutr. Soil Sci.* 157:459–465.
- Wu, J., M.E. Bauer, D. Wang, and S.M. Manson. 2008. A comparison of illumination geometry-based methods for topographic correction of QuickBird images of an undulant area. *ISPRS J. Photogramm. Remote Sens.* 63:223–236.
- Zieba-Palus, J. 1997. Application of transmittance and reflectance FT-IR microscopy to examination of paints transferred onto fabrics. *Mikrochim. Acta* 14:361–362.
- Ziechmann, W. 1980. *Huminstoffe*. (In German.) Verlag Chemie, Weinheim, Germany.